03/26/2026

Introduce $\vartheta_k = L t_k$ $\qquad \vartheta_k \in]0,2[$

$$f(x^{k+1}) \leq f(x^k) - \frac{1}{2L}\underbrace{\vartheta_k(2-\vartheta_k)}\|\nabla f(x^k)\|^2$$



$\rightsquigarrow$ descent is maximized at $\vartheta_k = 1 \quad \Leftrightarrow \quad t_k = \frac{1}{L}$

But only w.r.t. the estimate: $f(x^{k+1}) \leq f(x^k) - \frac{1}{2L}\|\nabla f(x^k)\|^2$

Rewrite: $\quad f(x^{k+1}) - f(x^k) \leq -\frac{1}{2L}\vartheta_k(2-\vartheta_k)\|\nabla f(x^k)\|^2$

Sum this up from $m = 0, \dots, k-1$.

$$f(x^k) - f(x^0) = \sum_{m=0}^{k-1} f(x^{k+1}) - f(x^k) \leq -\frac{1}{2L}\sum_{m=0}^{k-1}\vartheta_m(2-\vartheta_m)\|\nabla f(x^m)\|^2$$

Assume that $f(x^k) \geq M \quad \forall k$, then

$$\frac{1}{2L}\sum_{m=0}^{k-1}\vartheta_m(2-\vartheta_m)\|\nabla f(x^m)\|^2 \leq f(x^0) - M < \infty.$$

Consequently, $\quad \sum_{m=0}^{\infty}\underbrace{\vartheta_m(2-\vartheta_m)}_{>0}\underbrace{\|\nabla f(x^m)\|^2}_{>0} \leq 2L(f(x^0) - M)$

It follows that $\quad \lim_{m\to\infty}\vartheta_m(2-\vartheta_m)\|\nabla f(x^m)\|^2 = 0.$

If $\quad \liminf_{k\to\infty} t_k > 0 \quad$ and $\quad \limsup_{k\to\infty} t_k < \frac{2}{L}$,

then $\quad \liminf_{k\to\infty}\vartheta_k(2-\vartheta_k) = c > 0.$

$$\Rightarrow \quad 0 \leq \lim_{k\to\infty} c\|\nabla f(x^k)\|^2 \leq \liminf_{k\to\infty}\vartheta_k(2-\vartheta_k)\|\nabla f(x^k)\|^2 = 0$$

Thus, $\lim_{k\to\infty} \|\nabla f(x^k)\|^2 = 0$.

If $(x^k)_k$ has a convergent subsequence, i.e., there are indices $k_1 < k_2 < k_3 < \dots$ such that $\lim_{\ell\to\infty} x^{k_\ell} = x^*$, then

$\lim_{\ell\to\infty} \|\nabla f(x^{k_\ell})\|^2 = 0$ and since $\|\nabla f(x^{k_\ell})\|^2 \xrightarrow[\ell\to\infty]{} \|\nabla f(x^*)\|^2$,

(continuity of ∇f)

it follows that $\nabla f(x^*) = 0$.

In summary, each convergent subsequence of $(x^k)_k$ converges to a stationary point of f.

In particular, if $(x^k)_k$ converges, then $\lim_{k\to\infty} x^k = x^*$ is a stationary point. ✓

Discussion of prerequisites:

- Lipschitz continuity of ∇f ⇝ strong, but achievable
  ⇝ can be weakened
- $0 < t_k < \frac{2}{L}$ ⇝ can be achieved if Lipschitz constant $L > 0$ is known ⇝ estimate for L is sufficient
- $\liminf_{k\to\infty} t_k > 0$ ⇝ achievable
- $\limsup_{k\to\infty} t_k < \frac{2}{L}$ ⇝ achievable if Lipschitz constant is known

- Lower bound of $f(x^k)$, i.e. $f(x^k) \geq M$.

  $\rightsquigarrow$ if f has a minimizer $\bar{x} \in \mathbb{R}^n$, then $M = f(\bar{x})$ is feasible

  $$\Rightarrow \sum_{m=0}^{\infty} \underbrace{\alpha_m (2 - \alpha_m)}_{>0} \underbrace{\|\nabla f(x^m)\|^2}_{>0} \leq 2L \left(f(x^0) - f(\bar{x}) \right)$$

  $\rightsquigarrow$ if $\inf_{x \in \mathbb{R}^n} f(x) \geq M$, then M is feasible.



$\rightsquigarrow$ lower bound via $(f(x_k))_k$



$\rightsquigarrow$ lower bound (global)

- Existence of convergent subsequences

  $\rightsquigarrow$ if f is coercive, i.e., $\|x\| \to \infty \Rightarrow f(x) \to \infty$,

  then there is at least one convergent subsequence

  $\Rightarrow$ a stationary point could be found

  (+ there exists a minimizer).

Theorem 1:

Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously diff'able, ∇f be L-Lipschitz continuous, $L > 0$, $(x^k)_k$ generated by gradient descent

$\liminf_{k \to \infty} t_k > 0$, $\limsup_{k \to \infty} t_k < \frac{2}{L}$ and

x^* a minimizer of f. Then, each convergent subsequence of $(x_k)_k$ has a stationary point as limit.

If f is coercive, then there exists a subsequence with stationary point as limit.

Remark:

$$k \sigma \min_{0 \le m \le k-1} \|\nabla f(x^m)\|^2 \le \sum_{m=0}^{k-1} \underbrace{\alpha_m(2-\alpha_m)}_{\ge \liminf_{k\to\infty} \alpha_m(2-\alpha_m) = \sigma > 0} \underbrace{\|\nabla f(x^m)\|^2}_{\ge \min_{0\le m\le k-1} \|\nabla f(x^m)\|^2} \le 2L\left(f(x^0) - f(\bar{x})\right)$$

So,

$$\min_{0 \le m \le k-1} \|\nabla f(x^m)\|^2 \le \frac{2L}{\sigma k}\left(f(x^0) - f(\bar{x})\right) = O\left(\frac{1}{k}\right)$$

$\Rightarrow$ tracking the iterate with the smallest squared gradient norm gives rate of $O(\frac{1}{k})$.

Step-size rules:

- Line search: for x^k, choose t_k such that it minimizes $f(x^k - t\nabla f(x^k))$ over $t > 0$.

  Suppose that this always gives $t_k < \infty$. Then:

  $$f(x^{k+1}) \le f\left(x^k - \tfrac{1}{L}\nabla f(x^k)\right) \le f(x^k) - \frac{1}{2L}\|\nabla f(x^k)\|^2$$

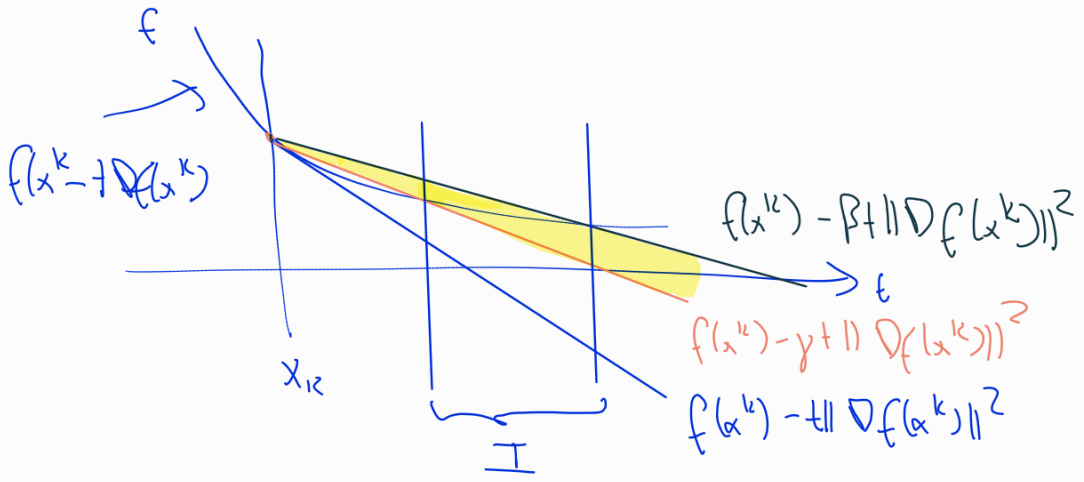
  $$\Rightarrow \sum_{k=0}^{\infty} \|\nabla f(x^k)\|^2 \le 2L\left(f(x^0) - f(\bar{x})\right)$$ if $\bar{x}$ is a minimizer

- Goldstein-rule: choose $t_k > 0$ such that

  $$f(x^k) - \gamma t_k \|\nabla f(x^k)\|^2 \le f(x^k - t_k \nabla f(x^k)) \le f(x^k) - \beta t_k \|\nabla f(x^k)\|^2$$

  for fixed $0 < \beta < \frac{1}{2} < \gamma < 1$.

Illustration:



I is the set of possible t according to the Goldstein rule

Interpretation of inequalities:

$f(x^k - t_k \nabla f(x^k)) \leq f(x^k) - \beta t_k \|\nabla f(x^k)\|^2$ $\rightsquigarrow$ appropriate descent

$\geq f(x^k) - \gamma t_k \|\nabla f(x^k)\|^2$ $\rightsquigarrow$ t_k not too small

$\rightsquigarrow$ some kind of convergence